

DEX Whale-Alert Bot

A production system that detects large wallet activity on decentralized exchanges in real time, filters it for genuine intent, and issues directional alerts — with every alert logged and backtested for signal quality. This document covers the architecture, the detection and filtering logic, the backtest methodology, and results over 1,199 resolved signals.

01 — EXECUTIVE SUMMARY

What it does, and what the numbers say

The system watches Binance-Futures-listed tokens for price momentum, then inspects the underlying DEX order flow to find wallets accumulating or distributing ahead of the crowd. Alerts are confirmed against price and market-cap conditions, delivered to Telegram, and tracked to resolution. The backtest below measures **signal quality** — whether an alert's direction was subsequently right — not realised trading P&L.

RESOLVED SIGNALS

1,199

4 chains · 6 types

STRONGEST CHAIN

+2.09 R

Solana signals

BEST AGGREGATE

+0.05 R

2% / 15% stop-take

UPTIME MODEL

24/7

VPS + edge workers

Headline: the edge is concentrated off-BSC. Solana, Ethereum and Base signals are clearly profitable, while BSC — which produces **92% of all signals** — sits at breakeven and dilutes the blended result. Rebalancing flow toward the smaller chains is the primary lever.

02 — ARCHITECTURE

A two-stage funnel across a split runtime

Detection runs on a dedicated VPS (its residential-grade egress is not rate-limited by the data providers), while confirmation, alerting and backtest bookkeeping run on Cloudflare's edge with a key-value store as the shared state. The scanner also relays prices into KV, so the edge never has to call a rate-limited provider directly.

STAGE 1 · WATCH

Binance Futures boards

Every 5 min: 5m / 30m / 24h gainers & losers rank the universe by momentum.



STAGE 2 · SELECT

DEX-pool mapping

Each mover mapped to its deepest pool; CEX-only tokens skipped, next-ranked backfills.



STAGE 3 · DETECT

Wallet order flow

Top buyers / sellers over a 60-min window trigger jump, step-up or activity signals.



STAGE 4 · CONFIRM

Edge worker + KV

Direction gate + price / market-cap checks promote a candidate to an alert.



STAGE 5 · LOG

Backtest & deliver

Telegram alert sent; price path recorded and replayed against a stop-take grid.

Design note. The split runtime was not the original plan — it is the answer to a real constraint (the edge provider's shared IP is blocked by the DEX data API). Moving detection to an owned IP and keeping the edge for stateful bookkeeping preserved 24/7 operation without paid data tiers.

03 — SIGNAL TAXONOMY

Two axes: how a token qualifies, and what the wallet did

A token first earns attention through one of three **entry timeframes** (the momentum board it topped), then a wallet inside it triggers one of three **detection types**. Both axes carry buy / sell direction.

Jump

A wallet bursts into the top-4 by volume having been outside the top-10 an hour earlier — a fresh large participant.

Step-up

A wallet in ranks 5–10 climbs at least two positions in an hour — steady accumulation or distribution.

Activity

A top-5 wallet trading heavily: ≥ 3 trades over \$10k, > 10 trades, or a single position over \$50k.

04 — QUALITY FILTERS

Three gates between a detection and an alert

- **Minimum size.** A wallet needs $\geq \$5,000$ of volume in the window and $\geq \$200$ per trade — noise and dust never reach the candidate stage.
- **Anti-wash.** If a wallet both bought and sold the same token in the window (opposite-side volume $\geq 25\%$ of the signal side), it is discarded — this removes volume-painting bots, which dominate BSC.
- **Direction gate.** A candidate can only confirm while price agrees with the signal — a sell never fires on a rising price. A counter-move beyond 2% invalidates the candidate outright rather than letting it wait out the move.

Confirmation itself requires either the wallet's volume to exceed 0.1% of live market cap, or price to move $\geq 3\%$ in the signal's direction. The executing wallet is also reconciled against the trade receipt, so the alert reports the **true token holder** when the operator address differs from it — a common bot pattern.

05 — BACKTEST METHODOLOGY

Record the path, replay the grid

Every confirmed alert opens a tracked position. Its price is sampled into a trail until it reaches $\pm 15\%$ or 48 hours. At close, that trail is replayed against a 42-cell grid of stop-loss (2–10%) and take-profit (2–15%) levels, so any risk-reward configuration can be evaluated after the fact from the same underlying data — no re-collection, no look-ahead.

- **Expectancy (R)** — average profit or loss per signal in units of risk; one R is the stop distance. A positive figure means the average signal made money at that configuration.
- **Win rate** — share of resolved signals that reached take-profit before stop-loss.
- **Best risk-reward** — the stop-take pair with the highest expectancy for a given segment.

Performance by network

Expectancy shown at each chain's best risk-reward. This is the report's central result.

NETWORK	SIGNALS	BEST RISK-REWARD	WIN RATE	EXPECTANCY
 Solana	52	2% / 15% (1 : 7.5)	36.4%	+2.09 R
 Base	16	3% / 5% (1 : 1.7)	71.4%	+0.90 R
 Ethereum	27	2% / 15% (1 : 7.5)	18.5%	+0.57 R
 BSC 92% of flow	1,104	2% / 10% (1 : 5)	16.0%	-0.04 R

By entry timeframe

BOARD	N	WIN RATE
24-hour	10	30.0%
30-minute	951	27.1%
5-minute	225	22.9%

By signal type

TYPE	N	WIN RATE
Step-up · buy	53	35.8%
Activity · buy	343	28.9%
Activity · sell	343	25.9%
Jump · buy	197	24.0%
Jump · sell	219	22.9%
Step-up · sell	44	18.6%

Risk-reward grid — aggregate

STOP / TAKE	RATIO	WIN RATE	EXPECTANCY
2% / 15%	1 : 7.5	12.3%	+0.05 R
2% / 10%	1 : 5	17.0%	+0.02 R
2% / 7%	1 : 3.5	21.9%	-0.01 R
3% / 10%	1 : 3.3	21.3%	-0.08 R
3% / 7%	1 : 2.3	27.0%	-0.10 R
5% / 10%	1 : 2	29.4%	-0.12 R

Reading the smaller samples. Solana (52), Ethereum (27) and Base (16) are early samples, and each chain's best risk-reward is chosen in-sample, so treat the exact figures as directional. The direction is consistent across all three non-BSC chains, and Solana's sample is now large enough to stand on its own.

07 — FINDINGS

What the data actually argues

- **Chain, not configuration, is the dominant variable.** No stop-take pair rescues the blended result, because the aggregate is 92% BSC. Split by chain, the picture inverts: three of four chains are clearly positive.
- **BSC flow is largely manufactured.** The anti-wash filter and the operator-vs-holder reconciliation both surfaced the same thing — much of BSC "whale activity" is volume-painting by bots, which is why its win rate collapses despite the largest sample.
- **Buy beats sell across every type.** Shorting momentum tokens fights continuation; every sell variant underperforms its buy counterpart.
- **Faster boards are not better.** The 5-minute board has the lowest win rate — entering after a 5-minute spike buys the top; slower boards give a cleaner entry.
- **The payoff shape is tight-stop / wide-take.** These are momentum tokens: they either continue far or reverse fast, so a small stop with a large target carries the expectancy, and symmetric configurations lose.

08 — ITERATION LOG

How the system reached its current form

● **Market-cap selection — abandoned**

The first version ranked tokens by market cap. Large caps trade on centralized venues; their DEX pools were near-dead and produced almost no signals. This forced a full rebuild.

● **DEX-volume selection + futures filter**

Rebuilt around the deepest pools by 24-hour DEX volume, restricted to tokens with Binance-Futures listings — a universe that actually carries informed flow.

● **Edge-IP block → split runtime**

The DEX data provider rate-limited the edge platform's shared egress IP. Detection moved to an owned VPS; the edge kept confirmation, alerting and state — and gained a price-relay so it never calls a blocked provider.

● **Static list → dynamic momentum boards**

A fixed token list became live 5m / 30m / 24h leaderboards, so the system now follows whatever is actually moving rather than a stale list.

● **Provider auth change, recovered**

A data provider began requiring request headers and silently rejected calls, freezing confirmations. Root-caused via persisted counters and repaired the same day; the price-relay now insulates the pipeline from provider-side limits.

● **Solana coverage + holder reconciliation**

Added Solana. Investigated why reported balances were often zero and found operator addresses differ from token holders — now parsed from the trade receipt and shown in the alert.

● **Anti-wash + direction gate + clean backtest**

Added the wash filter and the direction gate, then reset the backtest epoch to measure the tightened system cleanly against a 42-cell risk-reward grid.

09 — ROADMAP

Where the evidence points next

- **Rebalance flow off BSC.** Prioritise Solana, Ethereum and Base in board selection, or hold BSC to stricter thresholds, so capacity is spent where the edge lives.
- **Authenticity scoring.** Extend the operator-vs-holder reconciliation into a detection-time filter — genuine wallets (operator = holder) over bot infrastructure.

- **Grow the non-BSC sample.** Deepen Solana pool coverage to move its result from “directional” to “confirmed”.
- **On-demand reporting.** Fold this backtest into an incremental in-KV aggregate feeding a daily digest, instead of recomputing from full history each call.

Expectancy (R) — average P/L per signal in units of risk;
1 R = the stop distance. Positive is profitable.

Signal quality — this backtest measures directional correctness of alerts, not realised trading P&L.

Win rate — share of resolved signals reaching take-profit before stop-loss.

Method — per-alert price trail replayed against a 42-cell stop-loss / take-profit grid.

Sample: 1,199 resolved signals Universe: Binance-Futures-listed tokens · Ethereum · Base · BSC · Solana

Stack: VPS scanner + edge workers + KV state